Q.No	Question	Details
Q1	Types of Security Issues	Explain the following types of database security issues: 1. Legal and Ethical Issues 2. Policy Issues 3. System-related Issues
Q2	Database Security Control Techniques	Explain the following database security control techniques: 1. Access Control 2. Inference Control 3. Flow Control 4. Data Encryption
Q3	SQL View Creation	Consider a table Employee(EmpID, EmpName, Salary, DeptID). Write an SQL query to create a view that displays only EmpID, EmpName, and DeptID from the Employee table.
Q4	SQL Problem (LeetCode 570)	Solve the SQL problem: Managers with at Least 5 Direct Reports. Link: https://leetcode.com/problems/managers-with-at-least-5-direct-reports/
Q5	SQL Problem (LeetCode 577)	Solve the SQL problem: Employee Bonus. Link: https://leetcode.com/problems/employee-bonus/description/

DBMS Assignment

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Ans1.> Database security issues refer to problems, risks or challenges that threaten the confidentionality, integrity, and availability of data stored in a database system. These issues arise due to legal regulations, organizational policies or technical weakness in the system.

1.> Legal and ethical issues
- Organizations store sencitive data such as personal details, financial information and medical records.
- Employees must maintain confidentiality and responsible data usage.

2.> Policy Issues
- security policies define who can access, modify or delite data.
- weak policies can lead to unauthorized access and misuse of infomation.

3.> System - related issues
- Poor system configuration can allow hackers to exploit vulnerabilities.
- weak authentication, outdated software or lack of encryption can compromise database security.

Ans2.> Database security control Techniques

1> Access control
- It ensures that users can access only the data they are permitted to view or modify.
- Implemented user authentication, passwords and role based permissions.

2.> Inference control
- Sometimes user combine different pieces of data to infer sensitive information.
- Inference control techniques limit or modify query results to prevent such deductions.

3.) Flow Control

- Ensures that information moves only according to predefined security rules.
- Prevents sensitive data from flowing from high-security levels to lower-security levels.

4.) Data Encryption

- Only users with the correct decryption key can read the data.
- Protects data during storage & transmission.

Ans3)

```
CREATE VIEW Employee_View AS
SELECT EmpID, EmpName, DeptName.
FROM Employee;
```

Explanation

CREATE VIEW Employee_VIEW → Creates a virtual table named Employee_View.

SELECT EmpID, EmpName, DepID → Selects only these three columns.

FROM Employee → Data is taken from the Employee table.